Intro to Data Science HW 8

Copyright 2025, Jeffrey Stanton, Jeffrey Saltz, Christopher Dunham, and Jasmina Tacheva

```
# Enter your name here: Mars Schrag
```

USE OF ARTIFICIAL INTELLIGENCE: While you are allowed to use generative AI as a learning tool for this assignment (e.g., to help with brainstorming, to see how to code a certain concept, to help identify flaws in reasoning, or to spot confusing or underdeveloped code segments), it must not be used to input homework questions, problems, or instructions directly, and you may not copy and paste any AI-generated content into your work. By submitting this assignment, you confirm that all the work is your own.

Attribution statement: (choose only one and delete the rest)

```
# 1. I did this homework by myself, with help from the book and the professor.
# 4. I did this homework with help from ChatGPT or another generative AI tool,
# but I followed the homework guidelines.
```

This homework assignment introduces linear predictive modeling through the use of multiple regression, which is a tool for assessing the relationship between a set of predictor variables and an outcome variable. The term "multiple regression" originated from an early scientific observation of a phenomenon called "regression to the mean," but today it is simply a name for using linear modeling to analyze data.

In this homework, we will be predicting house prices using several predictor variables. To do so, we will use the publicly available "House Sales in King County, USA" dataset, which can be found at the following URL:

https://data-science-intro.s3.us-east-2.amazonaws.com/house-prices.csv

To begin, load the dataset into a pandas dataframe named "house_df". Next, use the relevant function(s) to generate a summary of the data. Add a comment explaining the number of variables and observations in the dataframe and show the function(s) you used to generate this information.

```r
house_df <- read.csv("https://data-science-intro.s3.us-east-2.amazonaws.com/house-prices.csv")
summary(house_df)
```

```
##       Home            Price             SqFt         Bedrooms
##  Min.   :  1.00   Min.   : 69100   Min.   :1450   Min.   :2.000
##  1st Qu.: 32.75   1st Qu.:111325   1st Qu.:1880   1st Qu.:3.000
##  Median : 64.50   Median :125950   Median :2000   Median :3.000
##  Mean   : 64.50   Mean   :130427   Mean   :2001   Mean   :3.023
##  3rd Qu.: 96.25   3rd Qu.:148250   3rd Qu.:2140   3rd Qu.:3.000
##  Max.   :128.00   Max.   :211200   Max.   :2590   Max.   :5.000
##    Bathrooms         Offers         Brick           Neighborhood
##  Min.   :2.000   Min.   :1.000   Length:128         Length:128
##  1st Qu.:2.000   1st Qu.:2.000   Class :character   Class :character
##  Median :2.000   Median :3.000   Mode  :character   Mode  :character
##  Mean   :2.445   Mean   :2.578
##  3rd Qu.:3.000   3rd Qu.:3.000
##  Max.   :4.000   Max.   :6.000
```

```r
dim(house_df)
```

```
## [1] 128   8
```

```r
# 128 observations, 8 variables
```

B. In the subsequent analysis, **Price** will serve as the outcome variable, while the size (in square feet), number of bedrooms and bathrooms, and whether the house is made of brick or not will serve as predictors. Include a comment briefly describing the outcome variable.

Include a second comment to explain the difference between the outcome variable and the predictors in linear regression.

```r
# Comment1: Price is the outcome variable because it will determine worth of
# the house
```

```r
# Comment2: The outcome is the final worth, while predictors are things that
# could determine the worth
```

C. Before proceeding with the analysis, it is essential to inspect the outcome and predictor variables for any missing values. Show the code you used to check for missing values in these variables.

Explain the output of the code in a comment - what does it say about the number of missing values in each variable?

```r
sum(is.na(house_df$Price))
```

```
## [1] 0
```

```r
sum(is.na(house_df$Bedrooms))
```

```
## [1] 0
```

```r
sum(is.na(house_df$Bathrooms))
```

```
## [1] 0
```

```r
sum(is.na(house_df$Brick))
```

```
## [1] 0
```

```r
sum(is.na(house_df$SqFt))
```

```
## [1] 0
```

```r
# there are no missing values
```

D. Next, it's essential to ensure that all four predictor variables listed in Step B are coded as numerical values. If any predictor is not already in numerical form, it must be converted. The only exception is the "Brick" variable, which has only two values: "Yes" and "No." The code below demonstrates how to convert it into a binary format — be sure to include a comment explaining what the code does, and provide the code you used to check the variable types of the other predictors.

```r
house_df$Brick <- as.numeric(as.factor(house_df$Brick), levels = c(0, 1))

# Comment:It makes yes and no into numbericals equal to 0 and 1
```

```r
# Check the format of the rest of the predictors:
house_df$Bathrooms <- as.numeric(house_df$Bathrooms)
house_df$Bedrooms <- as.numeric(house_df$Bedrooms)
house_df$SqFt <- as.numeric(house_df$SqFt)
```

Once the conversion is complete, make sure to check again for any missing values. If any missing values are present, an appropriate strategy such as interpolation() from the **imputeTS** package we have used in previous HW assignments should be used.

```r
sum(is.na(house_df$Price))
```

```
## [1] 0
```

```r
sum(is.na(house_df$Bedrooms))
```

```
## [1] 0
```

```r
sum(is.na(house_df$Bathrooms))
```

```
## [1] 0
```

```r
sum(is.na(house_df$Brick))
```

```
## [1] 0
```

```r
sum(is.na(house_df$SqFt))
```

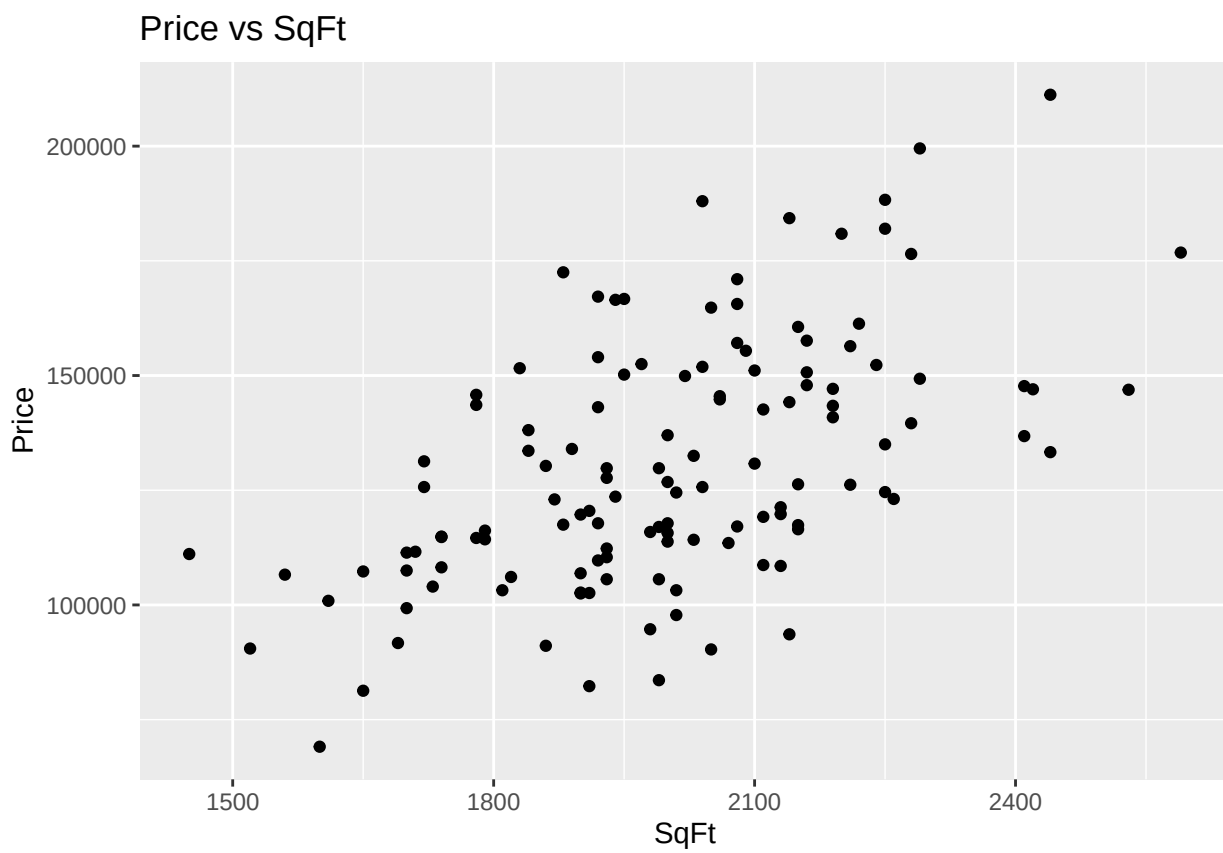
```
## [1] 0
```

E. To explore the relationship between each predictor and the outcome variable, we will create four visualizations using ggplot. In each plot, plot the outcome variable, price, on the Y-axis, and the respective predictor variable on the X-axis. It's up to you to decide on the most informative way of visualizing the relationship between the outcome variable and each predictor - e.g. by using scatterplots or boxplots. Don't forget to clearly label the X- and Y-axis using xlab() and ylab() and add a title to each plot.

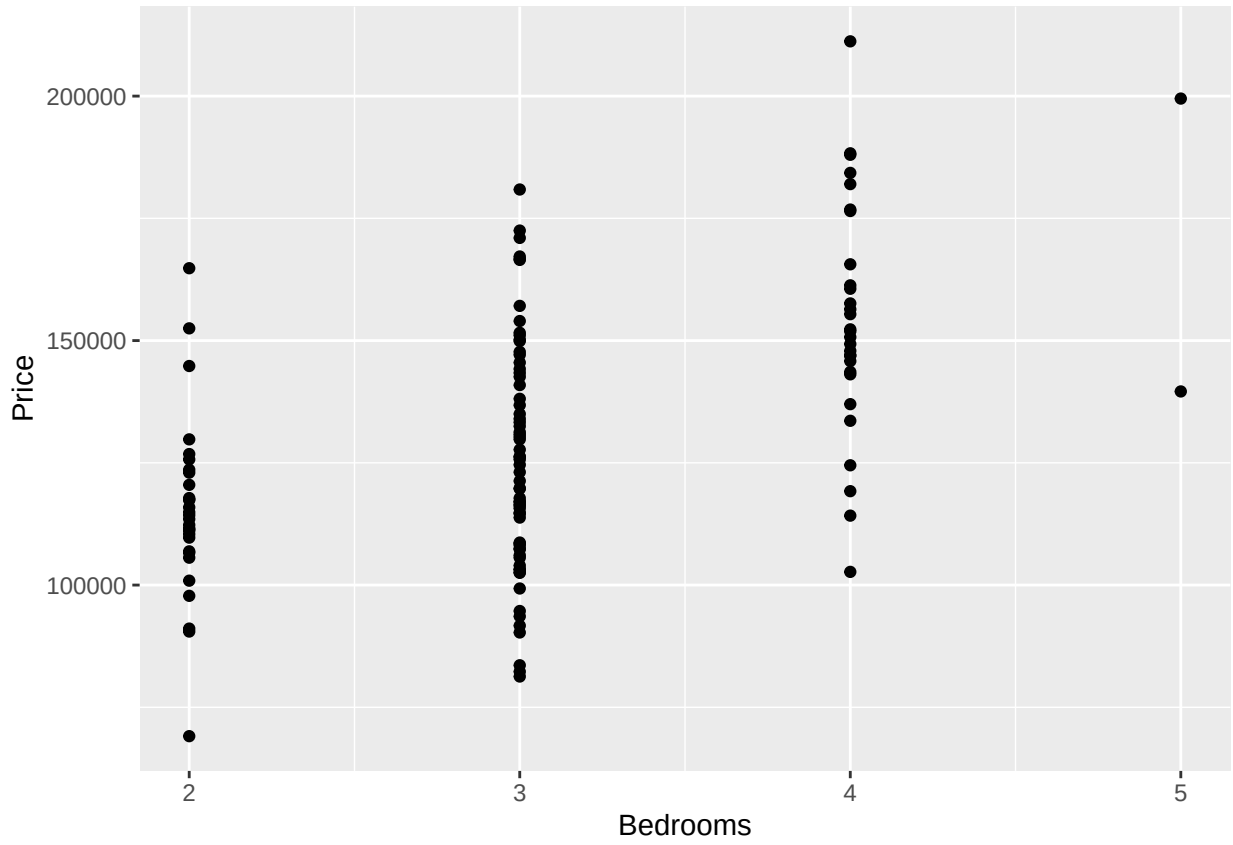
In a comment, explain your decision to use a particular kind of graph and describe each plot, indicating whether there appears to be a linear relationship between the outcome variable and the respective predictor variable.

```r
library(ggplot2)

ggplot(house_df, aes(x = SqFt, y = Price)) + geom_point() + xlab("SqFt") + ylab("Price") +
    ggtitle("Price vs SqFt")
```



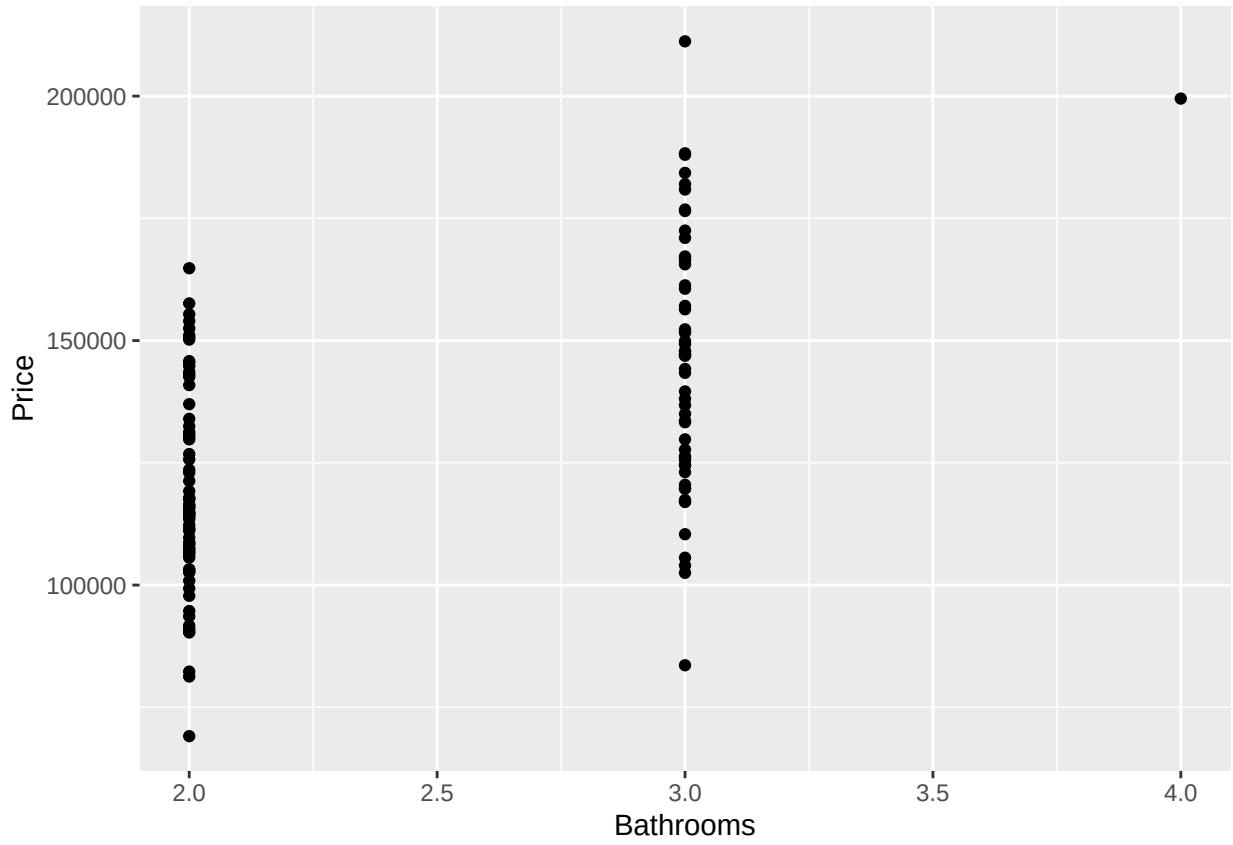

```r
ggplot(house_df, aes(x = Bedrooms, y = Price)) + geom_point()
```




```
xlab("Bedrooms") + ylab("Price") + ggtitle("Price vs Bedrooms")
```

```
## NULL
```

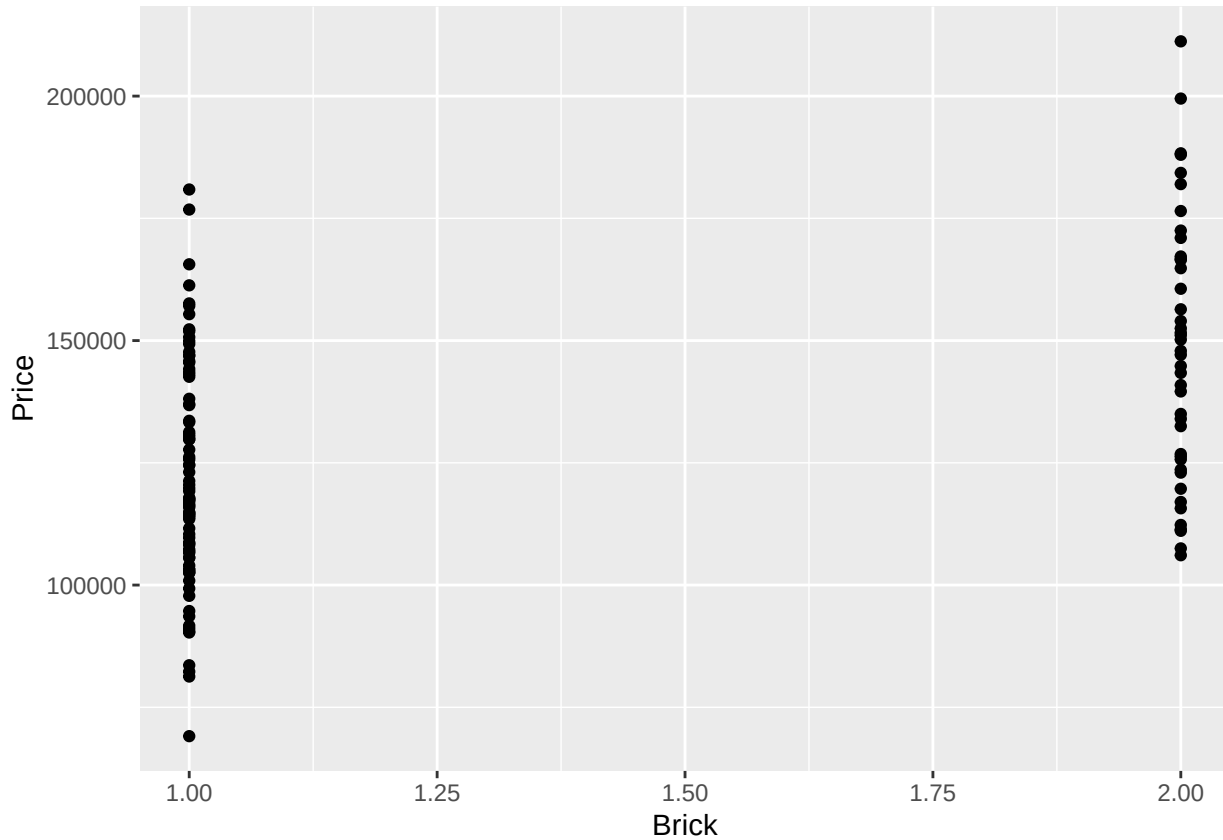
```
ggplot(house_df, aes(x = Bathrooms, y = Price)) + geom_point()
```




```
xlab("Bathrooms") + ylab("Price") + ggtitle("Price vs Bathrooms")
```

```
## NULL
```

```
ggplot(house_df, aes(x = Brick, y = Price)) + geom_point()
```




```r
xlab("Brick") + ylab("Price") + ggtitle("Price vs Brick")
```

```
## NULL
```

```r
# I chose scatterplot because it is the one I personally understand the most
```

F. Next, create a simple regression model predicting price based on size, using the lm( ) command. In a comment, report the coefficient (aka slope or beta weight) of size in the regression output, and if it is statistically significant, interpret it with respect to price. Report the adjusted R-squared of the model and try to explain what it means.

```r
model <- lm(Price ~ SqFt, data = house_df)
summary(model)
```

```
##
## Call:
## lm(formula = Price ~ SqFt, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -46593 -16644  -1610  15124  54829
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -10091.130  18966.104  -0.532    0.596
## SqFt            70.226      9.426   7.450  1.3e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22480 on 126 degrees of freedom
## Multiple R-squared:  0.3058, Adjusted R-squared:  0.3003
## F-statistic:  55.5 on 1 and 126 DF,  p-value: 1.302e-11
```

```r
# Comment: the size coefficient is 70.226, it is highly significant, and the
# r-squared is 30% so about 30% of price is explained by size
```

G. Create a multiple regression model predicting price based on size AND the number of Bedrooms. Ensure that both predictors are incorporated into a single model, rather than creating separate models for each predictor. This will allow us to assess the relationship between each predictor and the outcome variable while controlling for the effect of the other predictor.

```r
model_2 <- lm(Price ~ SqFt + Bedrooms, data = house_df)
summary(model_2)
```

```
##
## Call:
## lm(formula = Price ~ SqFt + Bedrooms, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -45993 -15402  -1446  11041  48352
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6367.60   17827.91  -0.357    0.722
## SqFt           49.50      10.11   4.895 2.97e-06 ***
## Bedrooms    12486.06    2947.13   4.237 4.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21100 on 125 degrees of freedom
## Multiple R-squared:  0.393,  Adjusted R-squared:  0.3832
## F-statistic: 40.46 on 2 and 125 DF,  p-value: 2.826e-14
```

H. Report the **coefficients** of each predictor that is statistically significant and interpret them in a comment.

How does the *adjusted R-squared** compare to the adjusted R-squared from Step F? Is this better or worse?

```r
# Comment: The bedrooms number is way higher than the sqft number. I believe
# this means bedrooms play a higher role in price than size does. The adjusted
# r-squared is 38%, which is higher than the previous one.
```

I. Use this code to create a one-row data frame with a single observation:

```r
predict_df <- data.frame(SqFt = 1200, Bedrooms = 3)
```

Apply the **predict( )** function to this data frame to predict the **expected value of price**:

```r
predict(model_2, predict_df)
```

```
##        1
## 90489.21
```

Can you assess the accuracy of the prediction using the model output from G? Explain in a comment.

```r
summary(model_2)
```

```
##
## Call:
## lm(formula = Price ~ SqFt + Bedrooms, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -45993 -15402  -1446  11041  48352
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6367.60   17827.91  -0.357    0.722
## SqFt           49.50      10.11   4.895 2.97e-06 ***
## Bedrooms    12486.06    2947.13   4.237 4.36e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21100 on 125 degrees of freedom
## Multiple R-squared:  0.393,  Adjusted R-squared:  0.3832
## F-statistic: 40.46 on 2 and 125 DF,  p-value: 2.826e-14
```

```r
# yes, the residual standard error shows how accurate the prediction is from
# just size and bedroom number
```

J. Create a new **multiple regression model**, with **Price** as the **outcome variable**, and **all four predictors** added to the model.

```r
full_model <- lm(Price ~ SqFt + Bedrooms + Bathrooms + Brick, data = house_df)
summary(full_model)
```

```
##
## Call:
## lm(formula = Price ~ SqFt + Bedrooms + Bathrooms + Brick, data = house_df)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -44438 -12879    618  14124  45344
##
```

```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -27188.542  15264.772  -1.781 0.077359 .
## SqFt            35.801      9.241   3.874 0.000173 ***
## Bedrooms     10873.060   2523.693   4.308 3.33e-05 ***
## Bathrooms     9818.430   3699.285   2.654 0.009002 **
## Brick        21909.107   3371.179   6.499 1.81e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17630 on 123 degrees of freedom
## Multiple R-squared:  0.5828, Adjusted R-squared:  0.5692
## F-statistic: 42.96 on 4 and 123 DF,  p-value: < 2.2e-16
```

K. **Analysis**: Interpret the significant coefficients of the new model you created and review the quality of the model by commenting on its **adjusted R-Squared**. Based on the work you have done so far, including the visualizations and regression models you created, do you think there is a relationship between price and the predictor variables?

```
# Comment: The adjusted r-squared is way higher at 56%, meaning all of the
# predictors account for 54% of the price outcome. There is still a large
# amount of error possible, but it is a better prediction with all four
# predictors. I think there is a relationship between price and predictor, and
# it can be seen using the data.
```